

SECTION 1 - MAGNET FIELD ADJUSTMENT / RAMPDOWN (FIELD DECREASE TO ZERO)



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.
- HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.

IMPORTANT !!!

Before starting Rampdown open Valve V3 to check for gas flow and exhaust plumbing cooling. No gas flow / cooling is an indication of ice in the Recondenser or exhaust plumbing which must be cleared before starting Rampdown. See REPLACEMENT / MAINTENANCE, Section 2, Ice Removal.

1-1 INTRODUCTION

The Main and Bucking Coils of the GE 0.7T OpenSpeed magnet are in a horizontal plane to generate a vertical magnetic field. Because of this configuration, it is essential that the magnet helium level remain between 95% and 98% during magnet field adjustment / rampdown so that helium stays in contact with the Top Bucking Coil. Failure to maintain this condition will result in a cooling loss of the Top Bucking Coil and could result in a quench.

This procedure maintains the required helium level during magnet field adjustment / rampdown through the controlled replenishment of liquid helium to compensate for loss through the Ramp Leads and Vent Valve V3.

1-1 INTRODUCTION (continued)

Because of the complexity and simultaneous tasks of magnet field adjustment / rampdown, two trained and certified persons are required to perform the steps. Person 1 performs the magnet field adjustment / ramp-down while person 2 simultaneously performs the fill.

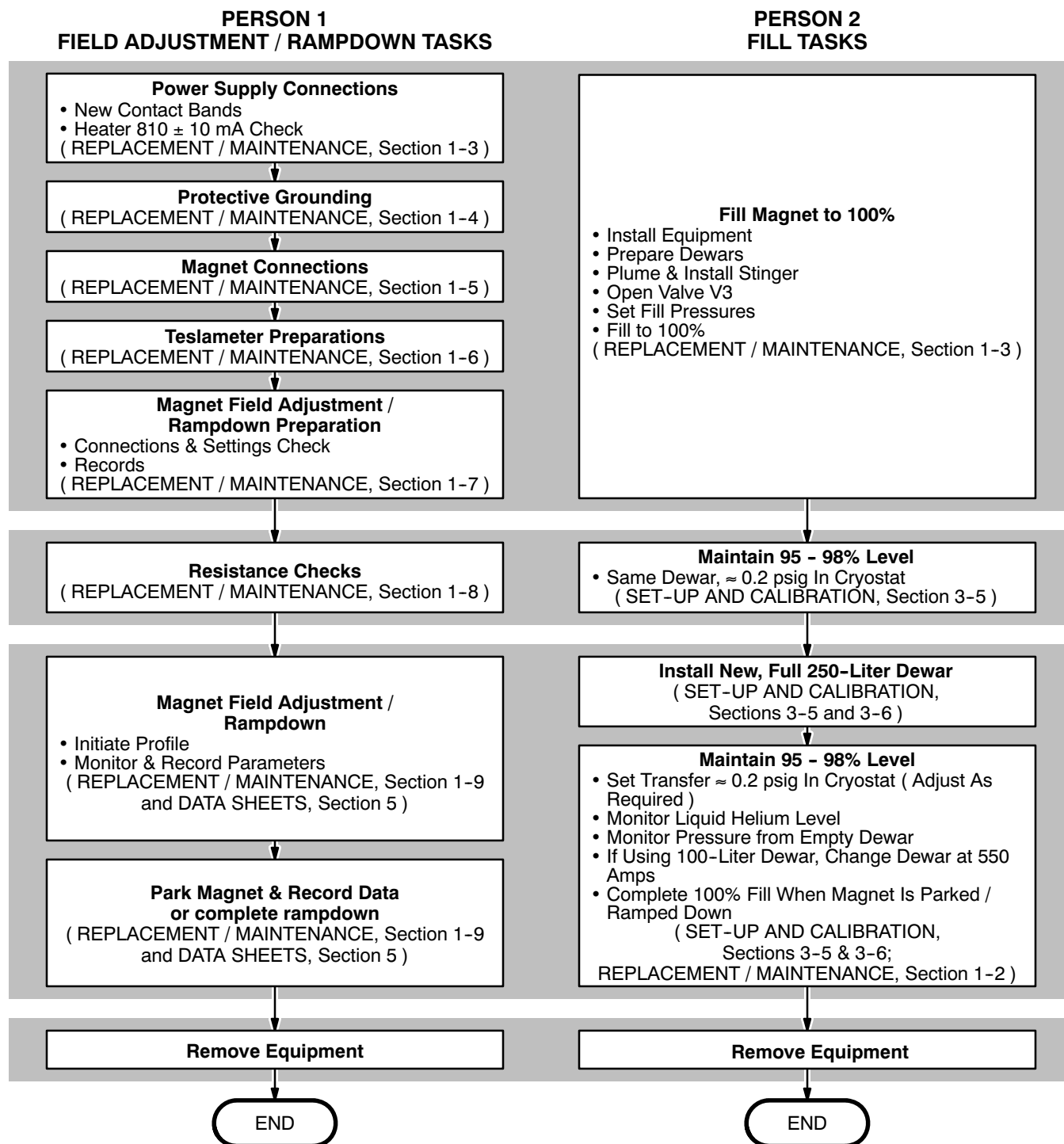
Before starting these procedures, review and fully understand all superconducting magnet portions of *2301164PRE, MR Magnet – Safety Requirements*, found in the SAFETY REQUIREMENTS tab of this manual. Fully comply with that document's Section 2-4, Magnet & Cryogen Service Safety Requirements. Have all "Work Assistants" or "Work Observers" comply with that document's Section 2-4, Buddy System Requirements & Certification.

Table 1-1 outlines the field adjustment / rampdown process steps. Illustration 1-1 outlines the tasks and sequence for each person. Complete DATA SHEETS, Table 5-2, Magnet Ramp / Fill Profile, for both ramp up and ramp down.

TABLE 1-1
RAMP / FILL PROCESS STEPS

Prerequisites
• Magnet Heaters must be stabilized. • Perform Power Supply checks to verify functional ramp supply. • Magnet Monitor must be functional and put into "FILL" mode.
The process
• Vent magnet down to .5 psi. • If the Upper Vessel helium level is < 80%, top off the magnet to at least 95%. Valve V3 should be open when helium level is greater than 70%. • If the Upper Vessel helium level is > 80%, proceed to next step. • Install Protective Grounding Circuit. • Insert the Main Lead Extensions. • Install the Hold-down Tool, but do not fully tighten. • Insert the fill line Stinger and begin filling the magnet. • Tighten the Hold-Down Tool. • When the Upper Vessel helium level is > 90%, do resistance checks. • When the resistance checks pass, change to a new 250-liter Dewar. • Park magnet at 29.791 MHz. This will account for the frequency drop when the supply is ramped down and the frequency increase when the magnet is shimmed. Specification for 0.7T OpenSpeed Magnet is 29.803 MHz $\pm$ 10 KHz. • Replace Power Supply Jumper Cable before removing the Main Lead Extensions. • After the supply is turned off, continue to fill while removing the Hold-Down Tool and Main Lead Extensions. This will give a higher helium level for magnet turnover to the customer. Hold-Down Tool and Main Lead Extensions should be removed quickly. • Put the Magnet Monitor back into normal operating mode.

1-1 INTRODUCTION (continued)



FIELD ADJUSTMENT / RAMPDOWN / FILL PROCESS
ILLUSTRATION 1-1

1-1 INTRODUCTION (continued)**IMPORTANT !!!**

It is **essential** that the following conditions are met to prevent a 250-liter helium Dewar from running dry during rampdown. Approximately 150 liters of helium will be used under these conditions.

- Follow the rampdown profile. Do not allow the ramp voltage to decay during each step of the ramp.
- Make sure contact resistance (< 150 mV @ 750 amps) is well within limits to minimize internal heat generation.
- Connect a new 250-liter helium Dewar after resistance checks (at the start of the magnet field adjustment / rampdown).
- Maintain helium level of 95 – 98% during the fill. Vessel pressure should be between 0.2 – 0.4 psig to maintain this helium level.

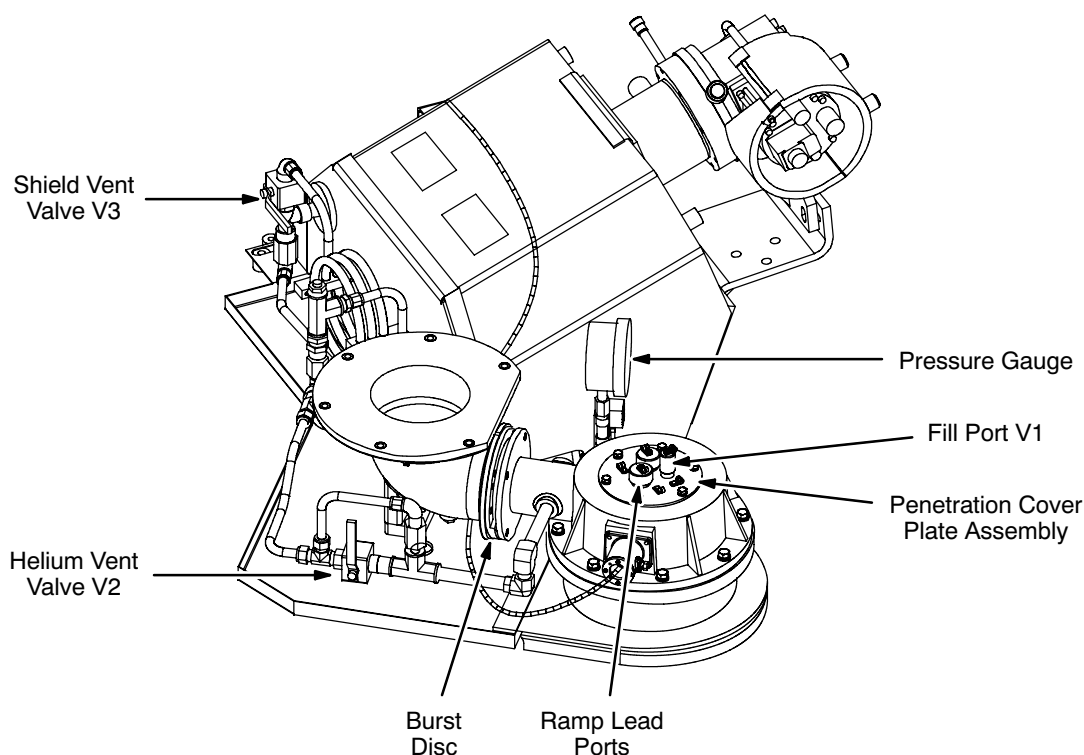
WARNING!

- **MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THE ODORLESS, INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE WILL DISPLACE OXYGEN AND CAUSE ASPHYXIATION UNLESS SAFELY VENTED AWAY FROM THE MAGNET ROOM.**
- **SECURE THE MAGNET ROOM DOORS IN THE OPEN POSITION BEFORE CHANGING MAGNETIC FIELD.**
- **REVIEW AND FOLLOW SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**

1-2 HELIUM FILL DURING FIELD ADJUSTMENT / RAMPDOWN - PERSON 2

Make sure the fill Stinger has been properly plumbed (venting liquid helium) before inserting into a ramped magnet to prevent any gas from entering the magnet and causing a quench.

1. Open Vent Valve V2 and vent magnet down to < 0.5 psig. Close V2. See Illustration 1-2.



V2 AND V3 VALVE LOCATIONS
ILLUSTRATION 1-2

2. If the helium level in the upper vessel is less than 90%, top the magnet to 90%.



Shield the pump-out port on the transfill line from cold helium gas that can harden the pump-out port o-ring and cause the line to go "soft."

3. Assemble the equipment and a new Dewar. Plume and insert the Stinger after the Ramp Lead Extensions and Hold-Down Fixture are installed.

1-2 HELIUM FILL DURING FIELD ADJUSTMENT / RAMPDOWN - PERSON 2 (continued)

4. Open Shield Coil Vent Valve V3. See Illustration 1-2. Fill the magnet to 98% in conformance with SET-UP AND CALIBRATION, Section 3-5, Helium Fill.

Note

Leave Valve V3 open sufficiently to maintain 0.2 psig helium transfer pressure during the entire ramping procedure to prevent pressure build-up.

5. Maintain a 95 – 98% helium level during resistance checks. Nominal cryostat pressure is 0.5 psig for maintaining this helium level.
6. Change to new, full 250 liter Dewar at the start of magnet field adjustment / rampdown in conformance with SET-UP AND CALIBRATION, Section 3-6, Changing Helium Dewars.

IMPORTANT !!!

It is **essential** that the magnet is at 98% liquid helium level and a new / full 250-liter Dewar is installed at the start of the ramp procedure to prevent the Dewar from running dry during the ramp.

7. Adjust Dewar pressure to produce ~ 0.5 psig cryostat pressure. Make minor adjustments to Dewar pressure as required to maintain between 95 – 98% liquid helium level during ramp.

CAUTION

Always maintain a positive pressure differential between the Dewar and the Cryostat to prevent any backflow of helium into the Dewar.

WARNING!

ALLOWING A HELIUM DEWAR TO EMPTY DURING RAMPING WILL CAUSE WARM GAS TO ENTER THE MAGNET AND CAUSE A QUENCH. CONTINUOUSLY MONITOR THE DEWAR AND MAGNET FOR A “WHISTLING” SOUND AND MAGNET PRESSURE FOR AN INCREASE; THESE INDICATE THE DEWAR IS EMPTY. WHEN AN EMPTY DEWAR IS SUSPECTED, IMMEDIATELY SHUT OFF THE DEWAR STINGER AND DISCONTINUE THE RAMP, MAINTAINING A CONSTANT RAMP CURRENT.

8. Continuously monitor the magnet for a noticeable pressure increase and the Dewar Stinger for a whistling sound indicating that the Dewar is empty. If the Dewar is empty, immediately shut off the Dewar Stinger and change Dewars in conformance with SET-UP AND CALIBRATION, Section 3-6, Changing Helium Dewars.
9. Once the magnet is ramped down and parked, make sure the magnet is at 90% helium fill level in the Top Helium Vessel. Then close V3 and immediately discontinue helium transfer. Remove the helium fill equipment in conformance with SET-UP AND CALIBRATION, Section 3-5, Liquid Helium Fill.

1-3 RAMP POWER SUPPLY CIRCUIT CONNECTIONS - PERSON 1

Introduction

At least once a year calibrate the Ramp Power Supply in conformance with FUNCTIONAL CHECKS, Section 8, ESS7.5-1000-2-D-1236 Main Power Supply Calibration / Check.

Make sure the Power Supply Cabinet is installed and checked out in conformance with the vendor manual supplied with the unit before making any power supply connections to the magnet. See INTRODUCTION, Section 2, Vendor Manual Matrix, for supplier manual numbers.

SCHEMATICS / INTERCONNECTS, Section 1-1, Magnet System Interconnects, shows the cable numbers and interconnect patterns covered in this Section.

The Main Coil Power Cables, Switch Heater Cables and Voltage Sense Leads are provided as part of the Ramp Cart / Cable Kit (1000 Amp) 2180589.

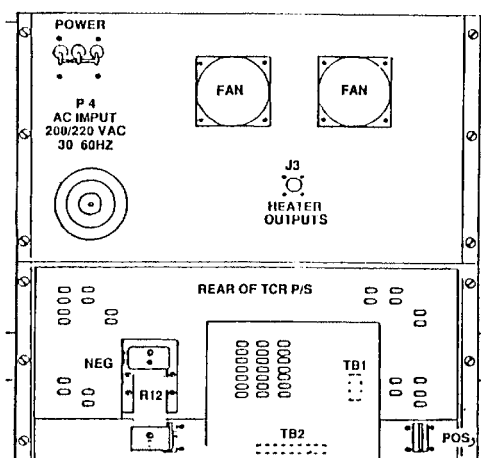
Use Main Lead Extensions 46-294204G1 from the Magnet Ramping Equipment Kit 46-260703G6.

Procedure

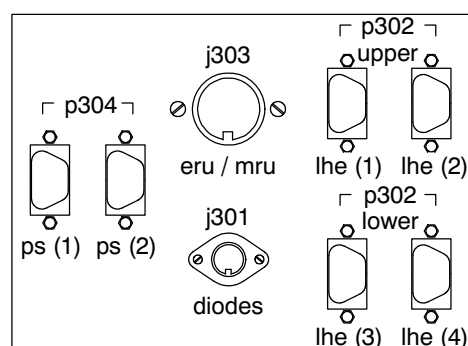
1. Verify that the input power to the Main Coil Power Supply is disconnected.
2. Connect "P3" on Switch Heater Cable 46-281667G1 to Heater Outputs "J3" on the rear of the Main Coil Power Supply Cabinet (MS6-A1). See Illustration 1-3.

Note:

This magnet incorporates a "B0 Coil" to decrease moving-metal sensitivity of the magnet field. The B0 Coil Heater is energized through the Ramp Power Supply's Axial Shim Heater Circuit. The Ramp Power Supply's Axial Shim Heater must remain on throughout the field adjustment / rampdown procedure in order to quench the B0 Coil field.



Power Supply (Rear)



Connector Panel

MAGNET POWER SUPPLY OUTPUT CONNECTIONS AND MAGNET CONNECTOR PANEL

ILLUSTRATION 1-3

1-3 RAMP POWER SUPPLY CIRCUIT CONNECTIONS - PERSON 1 (continued)

3. Connect "J304" on the Switch Heater Cable to "P304" on the Magnet Connector Assembly (MS1-A3, A1). See Illustration 1-3.

Note

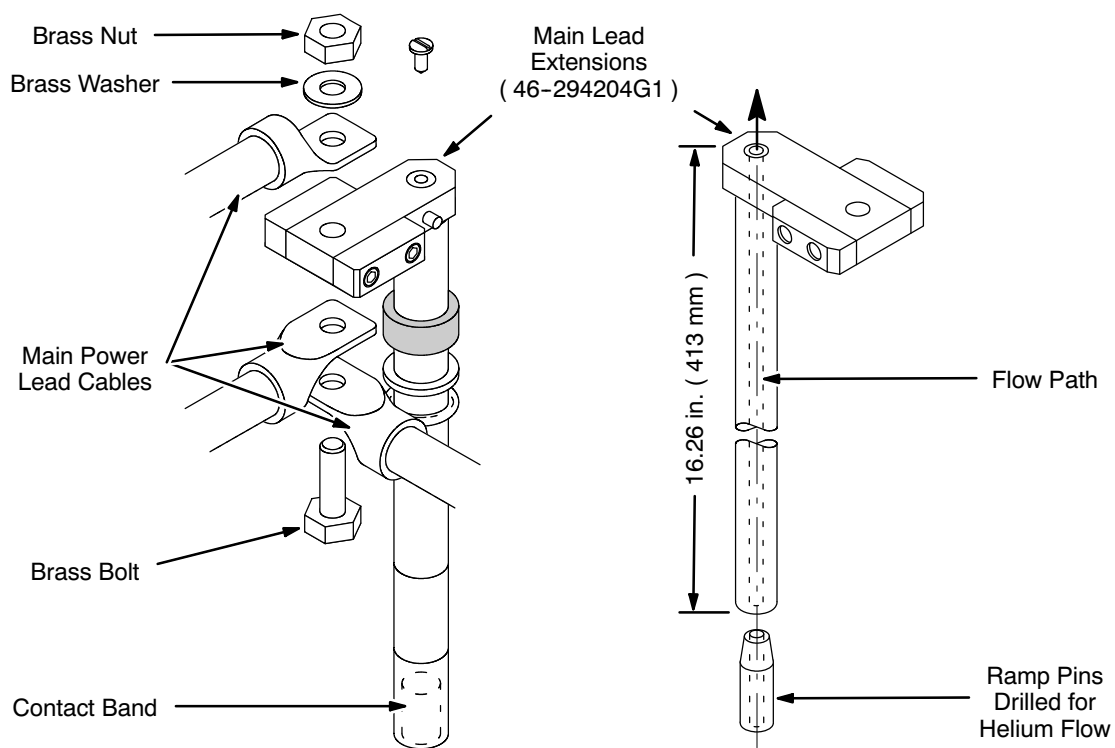
Make sure the nuts installed in Steps 4 and 6 below are tightened sufficiently to prevent a high resistance contact. Connect the red cables to the Positive (+) Buss Bar and the black cables to the Negative (-) Buss Bar on the Main Power Supply. For magnets ramped with normal polarity make sure the red (+) cables are connected to the "+" Magnet Ramp Port and the black (-) cables are connected to the "-" Magnet Ramp Port.

4. Connect the Main Coil Power Cables ("+" Red: 46-260723G1; "-" Black: 46-260723G2) to the Main Power Supply Buss Bars with the .375 inch brass nuts bolts and washers provided in the Magnet Ramping Equipment Kit (46-260703G4).

Note

Three pairs of Main Coil Power Cables must be used when ramping the GE 0.7T Open-Speed magnet.

5. Replace the Contact Bands on Main Lead Extension (46-294204G1) according to REPLACEMENT / MAINTENANCE, Section 8, Main Lead Extension Contact Band Replacement. Make sure the Gas Flow Holes are not blocked.
6. Connect the other end of the Main Power Cables to the Main Ramp Lead Extensions with the brass nuts, bolts and washers provided. See Illustrations 1-4 and 1-9.

1-3 RAMP POWER SUPPLY CIRCUIT CONNECTIONS - PERSON 1 (continued)

HARDWARE MOUNTING ON RAMP LEAD EXTENSIONS
ILLUSTRATION 1-4

1-4 PRE-RAMPDOWN PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1**WARNING!**

MAKE SURE THE POWER SUPPLY IS ALWAYS COMPLETELY GROUNDED IN CONFORMANCE WITH THE FOLLOWING PROCEDURE BEFORE INSERTING OR EXTRACTING ANY MAIN LEAD EXTENSION TO THE MAGNET (DURING RAMP UP, RAMP DOWN OR REPARKING MAGNET). GROUNDING WILL PREVENT THE REMOTE POSSIBILITY OF A HIGH VOLTAGE BETWEEN THE MAIN LEAD EXTENSION AND THE CRYOSTAT GROUND IF AN UNLIKELY QUENCH AND ARC / SHORT TO THE CRYOSTAT OCCUR WHEN THE MAIN LEAD EXTENSION IS INSERTED / EXTRACTED.

IMPORTANT !!!

Requirements during ramping:

- Always install the Pre-Ramp Protective Grounding Circuit and verify Main Lead Extension grounding to Cryostat ground before inserting or extracting any Main Lead Extension.
- Once the Main Lead Extensions are inserted, disconnect and remove the Power Supply Jumper Cable from the power supply while wearing non-absorbant leather gloves with NO holes / tears. This removes the short across the power supply and enables magnet ramping up / down.
- Once the magnet is parked and the power supply turned off — but before removing the Main Lead Extensions — reconnect the Power Supply Jumper Cable's twist connectors to the power supply while wearing non-absorbant leather gloves with NO holes / tears. This re-establishes full power supply grounding.
- After ramping up / down, when the power supply has been turned off and the Main Lead Extensions have been extracted, remove the Power Supply Jumper and Grounding Cables.

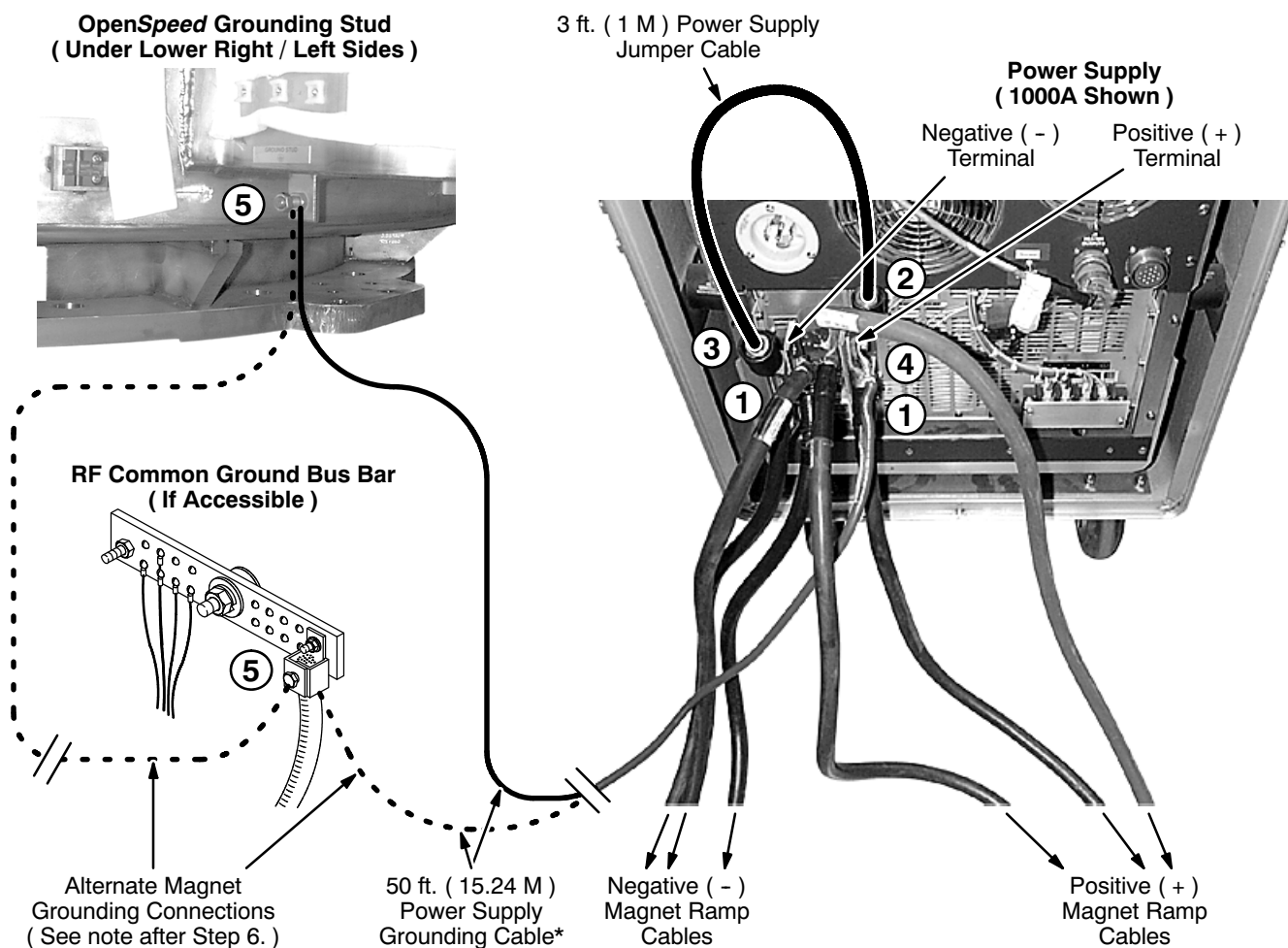
Equipment

- 3 ft. (1 M) Power Supply Jumper Cable
- 50 ft. (15.24 M) Power Supply Grounding Cable
- .375-16 x 1.75 inch brass screws, washers and nuts (2)

1-4 PRE-RAMPDOWN PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1 (continued)

Procedure

1. Connect the Magnet Ramp Leads to the Ramp Power Supply in the standard configuration and using the .375-16 x 1.75 inch brass screws, washers and nuts. See Illustration 1-5, label "1".

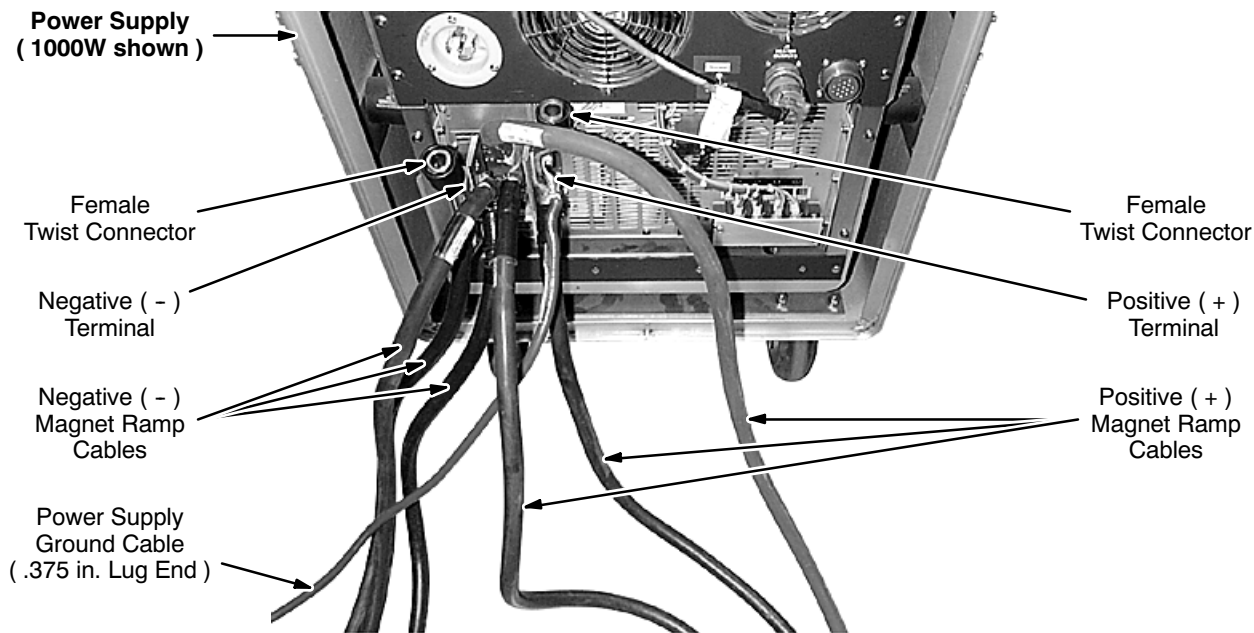


* 3/8 inch lug at Power Supply; .5 inch lug at Magnet.

PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS
ILLUSTRATION 1-5

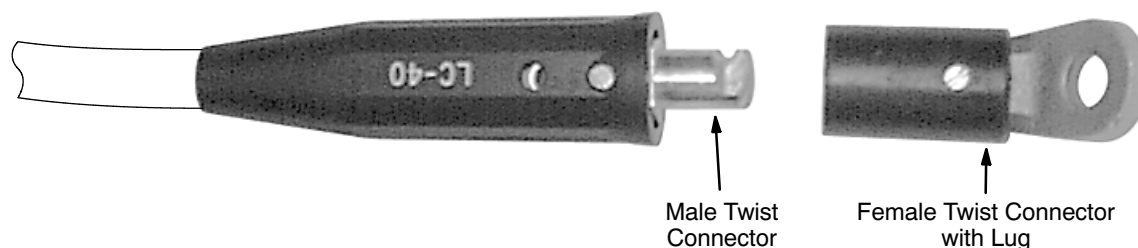
1-4 PRE-RAMPDOWN PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1 (continued)

2. Disconnect the female twist connector lugs from the 3 ft. (1 M) Power Supply Jumper Cable and then connect the lugs to the positive (+) and negative (-) power supply output terminals. See Illustration 1-6.



FEMALE TWIST CONNECTORS ATTACHED TO POWER SUPPLY OUTPUT TERMINALS
ILLUSTRATION 1-6

3. Connect one twist connector lug (shown in Illustration 1-7) of the Power Supply Jumper Cable to the power supply's positive (+) terminal. Make sure the twist connector is fully engaged. See Illustration 1-5, label "2".
4. Connect the other twist connector lug (shown in Illustration 1-7) of the Jumper Cable to the power supply's negative (-) terminal. Make sure the twist connector is fully engaged. See Illustration 1-5, label "3".
5. Connect the .375 inch (9.5 mm) lug of the 50 foot (15.24 M) Power Supply Grounding Cable to the power supply's positive (+) terminal. See Illustration 1-5, label "4".



TWIST CONNECTOR ON POWER SUPPLY JUMPER CABLE
ILLUSTRATION 1-7

1-4 PRE-RAMPDOWN PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1 (continued)

6. Connect the Grounding Cable's .5 inch (12.7 mm) lug to the magnet's grounding lug. See Illustration 1-5, label "5".

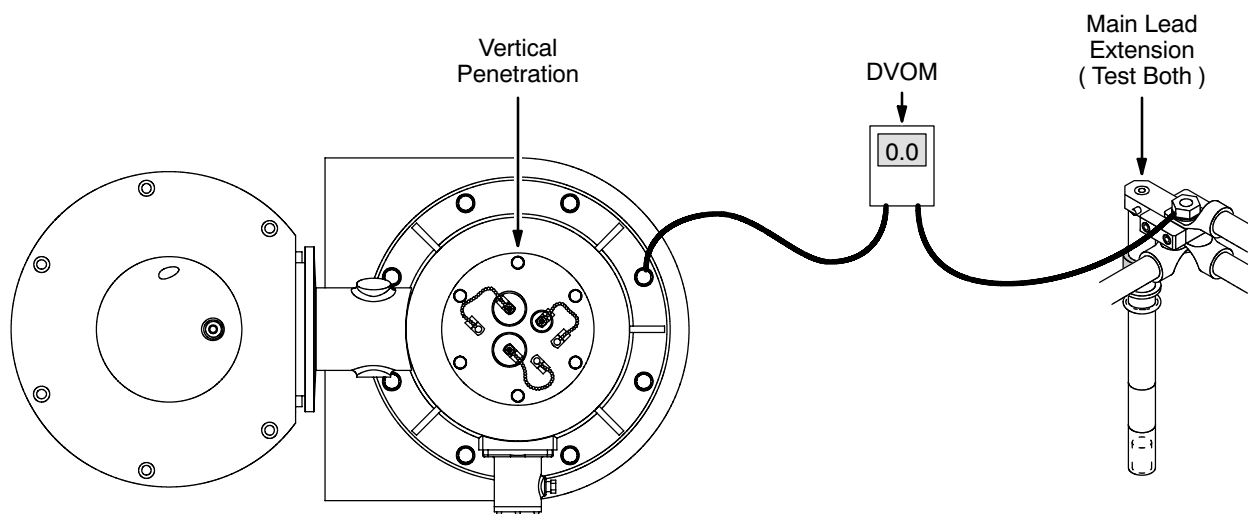
Note

If the magnet's grounding cable is connected to the RF Common Bus Bar at the Penetration Panel, the 50 foot (15.24 M) Power Supply Grounding Cable can be connected between the power supply and the magnet or PDU ground wire clamp on the RF Common Bus Bar if more convenient. If the magnet is not grounded to the Penetration Panel, the Grounding Cable **must** be connected to the Cryostat Grounding Stud. See Illustration 1-5.

CAUTION

Make sure there is a common ground (resistance = 0 ohms) between the Vacuum Vessel and the Main Lead Extension connections.

7. Verify that the resistance between the Vacuum Vessel at the Vertical Penetration and each Main Lead Extension connection is zero ohms as read on a DVOM. See Illustration 1-8.



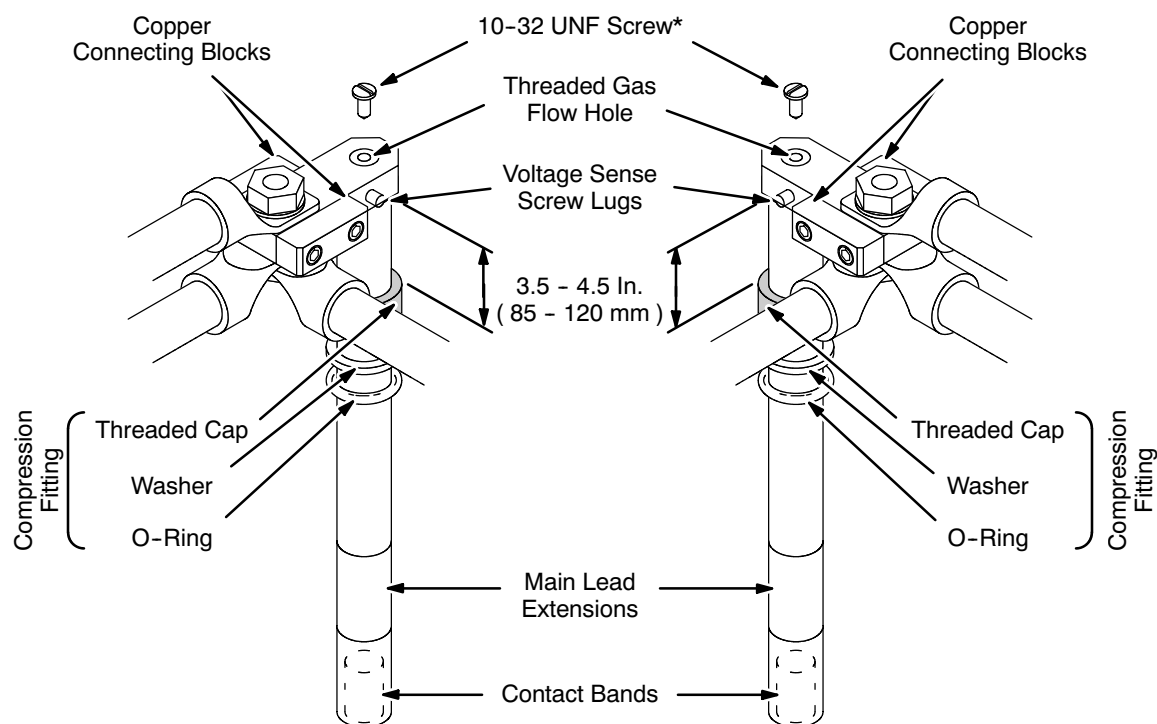
VACUUM VESSEL TO MAIN LEAD EXTENSION RESISTANCE CHECK
ILLUSTRATION 1-8

1-5 MAGNET CONNECTIONS - PERSON 1

1. Remove the threaded caps, washers and o-rings (46-294104P1, 46-294105P1, and 46-260389P1) from the plastic bag provided in the Magnet Ramping Equipment Kit (46-260703G4) and mount them on the Main Lead Extensions. See Illustration 1-9.

WARNING!

DO NOT LEAVE THE VENT SCREWS IN THE GAS FLOW HOLES FOR “DVM” CONNECTION SCREWS. THIS MAY CAUSE A QUENCH.



* Used to temporarily stop gas flow. Remove prior to ramping.

COMPRESSION FITTING MOUNTING ON MAIN LEAD EXTENSIONS
ILLUSTRATION 1-9

1-5 MAGNET CONNECTIONS - PERSON 1 (continued)

2. Place the end of the Main Power Cables on the Service Turret (MSI-A3, A3). Allow about 3 feet (900 mm) of slack.
3. Ty-wrap the cables at convenient locations on top of the magnet to avoid cable movement during ramping down.

WARNING!

- **MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THE ODORLESS, INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE WILL DISPLACE OXYGEN AND CAUSE AS-PHYXIATION UNLESS SAFELY VENTED AWAY FROM THE MAGNET ROOM.**
- **WAIT FOR THE LEAD EXTENSIONS TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE FULLY ENGAGING THEM, TO PREVENT THE RISK OF QUENCHING THE MAGNET.**
- **CONNECTION POLARITIES MUST BE THE SAME AS IN DATA SHEETS, SECTION 5, MAGNET RAMPING & PARKING CURRENT LOG, TO PREVENT THE POSSIBILITY OF MISWIRING AND A RESULTANT QUENCH DURING FUTURE RAMPING OF THE MAGNET.**
- **REVIEW AND FOLLOW SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**

WARNING!

THE CRYOSTAT EXHAUST SYSTEM HAS A 7.5 PSIG RELIEF VALVE, ALLOWING INTERNAL CRYOSTAT PRESSURE TO BUILD UP TO 7.5 PSIG. BEFORE REMOVING EITHER RAMP OR FILL CAPS OR LOOSENING ANY COMPONENT RESULTING IN THE RELEASE OF CRYOGEN PRESSURE:

- a. **OBSERVE THE INTERNAL CRYOSTAT PRESSURE GAUGE READING,**
- b. **TURN VESSEL PRESSURE ALARM OFF IN 'CONFIGURE / ALARMS' TAB**
- c. **SET MAGNET MONITOR PRESSURE TO 0.5 PSIG, AND**
- d. **VENT THE CRYOSTAT DOWN TO 0.5 PSIG THROUGH VENT VALVE V2.**

THIS WILL PREVENT EXPOSURE TO RAPID EXHAUSTING OF COLD HELIUM GAS DURING A SERVICE OPERATION.

4. Check the Cryostat Pressure Gauge. If the pressure is above 0.5 psig, slowly open Helium Vent Valve V2 and allow pressure to drop to 0.5 psig. Then close Valve V2.

1-5 MAGNET CONNECTIONS - PERSON 1 (continued)

WARNING!

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **MAKE ALL PRE-RAMP POWER SUPPLY PROTECTIVE CIRCUIT CONNECTIONS IN CONFORMANCE WITH SECTION 1-4, PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1.**
- **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL BEFORE CONTACTING THE MAIN LEAD PINS.**
- **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
- **WEAR NON-ABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
- **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**

CAUTION

To avoid the possibility of a quench, do not insert the Main Lead Extensions into the magnet unless the helium level is at 98% and then insert them slowly, with the 10-32 screws removed from Gas Flow Holes, to cool the Main Lead Extensions before insertion. Make sure gas is exiting Gas Flow Holes.

5. Verify that the cryostat pressure is < 0.5 psig.
6. Remove the cap on one Main Lead Extension receptacle located on the Vertical Penetration. Make sure the gasket inside the cap does not get lost.

CAUTION

Make sure the Gas Flow Holes in the Main Lead Extensions are not blocked and that gaseous helium is exiting through the holes. Blocked holes prevent heat dissipation and result in high contact resistance.

7. Insert one Main Lead Extension into the open receptacle slowly enough to allow the Lead Extension to cool during insertion.

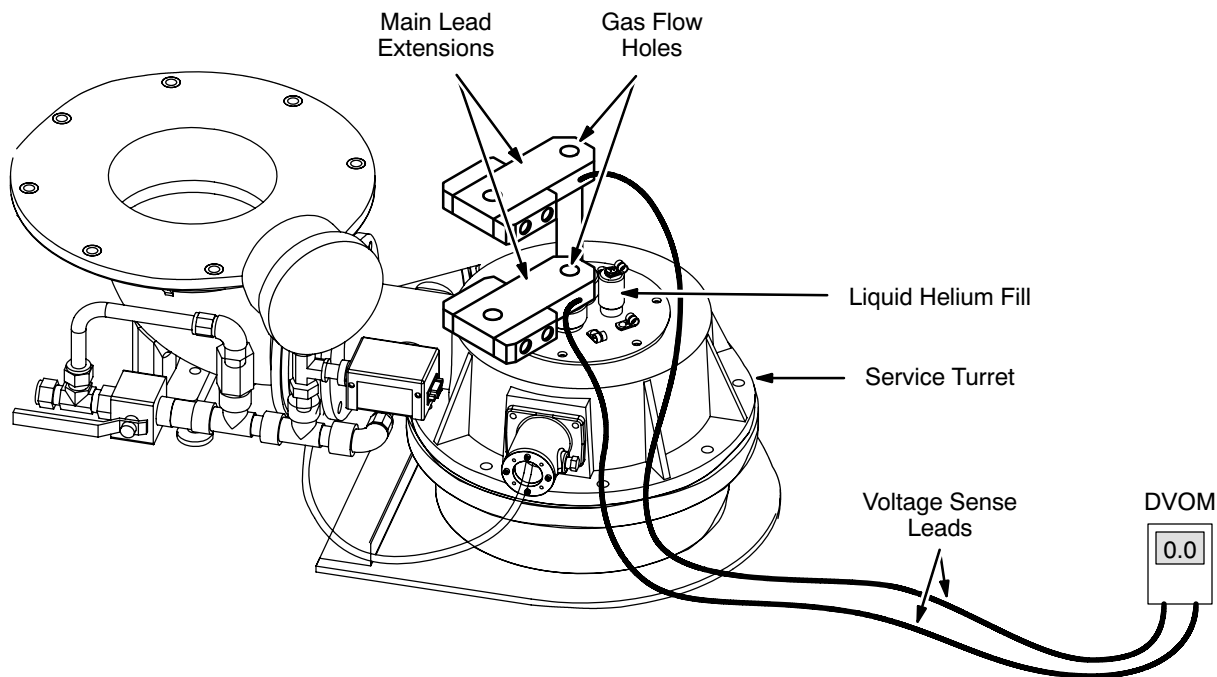
1-5 MAGNET CONNECTIONS - PERSON 1 (continued)

8. Continue inserting the Lead Extension until 85 – 120 mm (3.5 – 4.5 inches) is left exposed, then loosely tighten the compression fitting.
9. Repeat Steps 6 – 8 for the second Main Lead Extension.

Note

The Main Lead Extensions will depress approximately 1 inch (25 mm) from the point of contact to the fully engaged position. A firm contact will be felt when fully engaged. Do not rotate the Main Lead Extensions excessively when in the engaged position as internal contact wear could result.

10. When the Main Lead Extensions are sufficiently cooled, loosen the compression fittings and fully engage the Main Lead Extensions. Loosely tighten the compression fittings.
11. Retrieve the magnet polarity from DATA SHEETS, Section 5, Magnet Ramping & Parking Log.
12. Connect the Voltage Sense Leads to the Main Lead Extensions by clamping the spade lugs on the Voltage Sense Leads to the Main Lead Extensions with the screw lugs provided on the Lead Buss Bar. Terminate the other end of Voltage Sense Leads to a DVM or VOM placed near the Main Coil Power Supply. See Illustration 1-10.

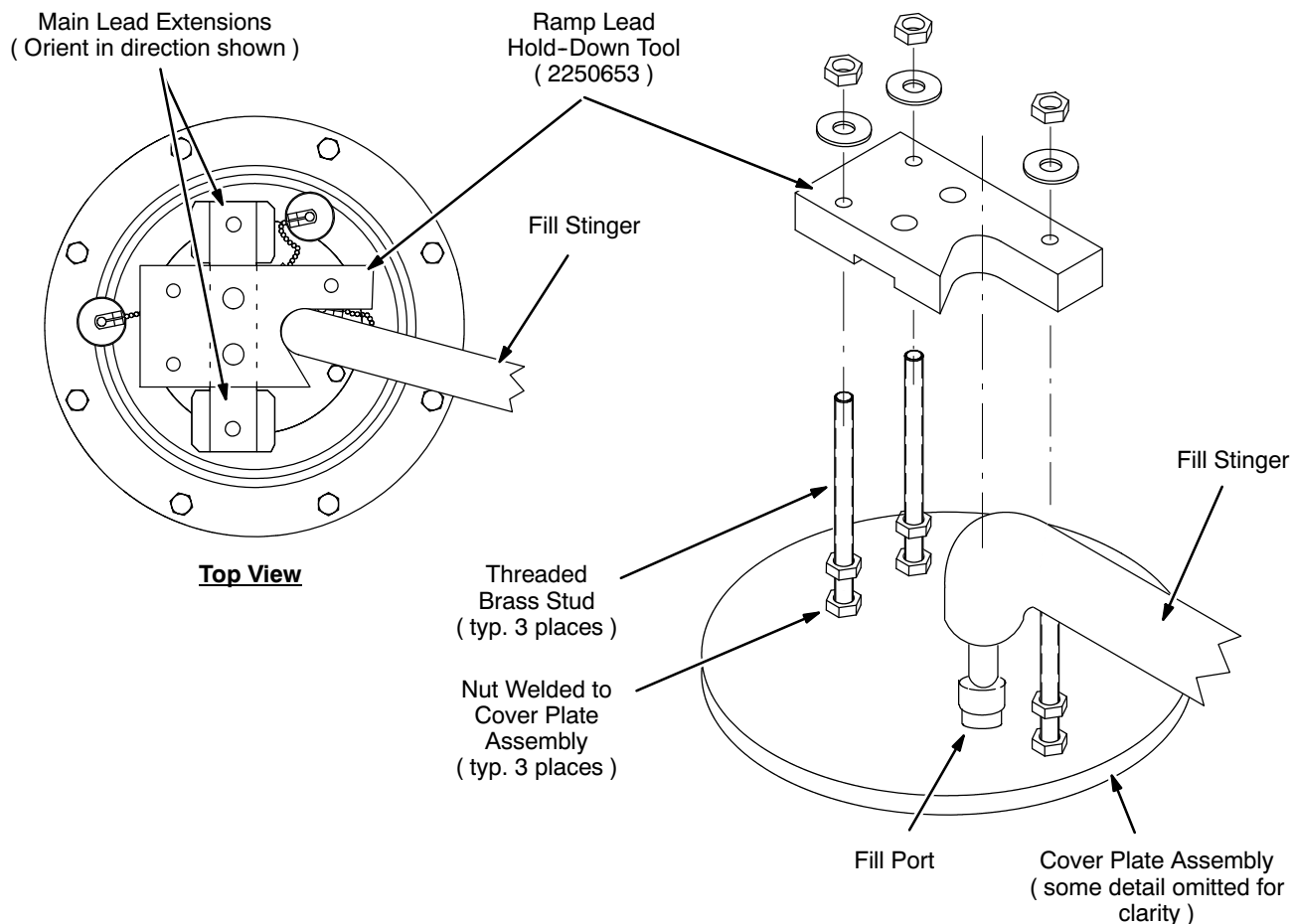


MAIN COIL AND VOLT SENSE LEAD CONNECTIONS
ILLUSTRATION 1-10

13. Insert three threaded brass studs into the nuts on the Vertical Penetration Cover Plate. See Illustration 1-11.

1-5 MAGNET CONNECTIONS - PERSON 1 (continued)

14. Install the Ramp Lead Hold-Down Tool (2250653). Make sure it is oriented properly, as shown in Illustration 1-11, to minimize Ramp Lead motion and help obtain good contact resistance.
15. Thread and tighten top nuts to hold the tool in place.
16. Hand tighten the Ramp Port compression nuts.



RAMP LEAD HOLD-DOWN TOOL
ILLUSTRATION 1-11



Make sure the Gas Flow Holes in the Main Lead Extensions are not blocked (i.e., the 10-32 screws have been removed) and that gaseous helium is exiting through the holes before ramping down. Blocked holes prevent heat dissipation and result in high contact resistance.

1-6 TESLAMETER / PROBE SET-UP AND ADJUSTMENT - PERSON 1

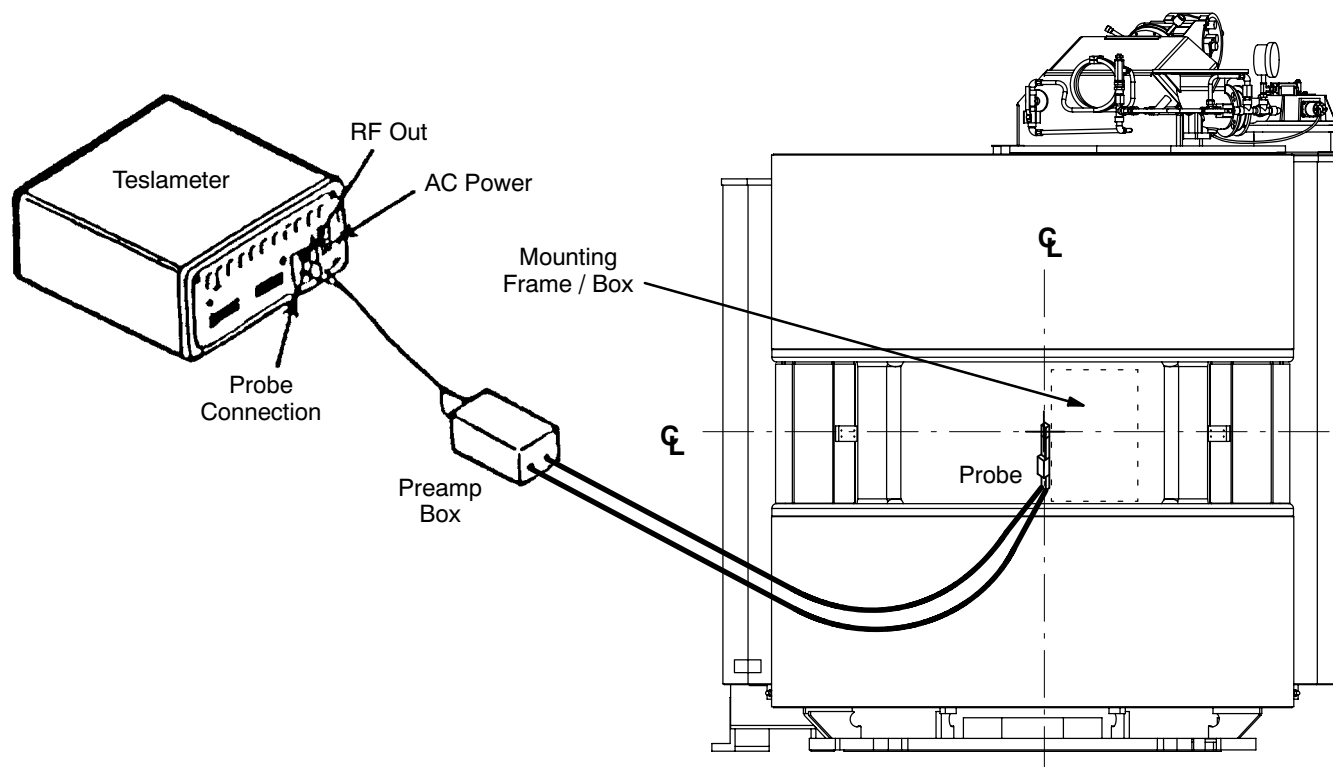
1-6-1 Set-Up

1. Mount and center the #4 Probe (0.35T – 1.05T) in the magnet using a cardboard / wood box or frame made from nonferromagnetic material. See Illustration 1-12.

Note

Centering the Probe between enclosure surfaces will give sufficiently accurate magnet field strength readings for ramping purposes.

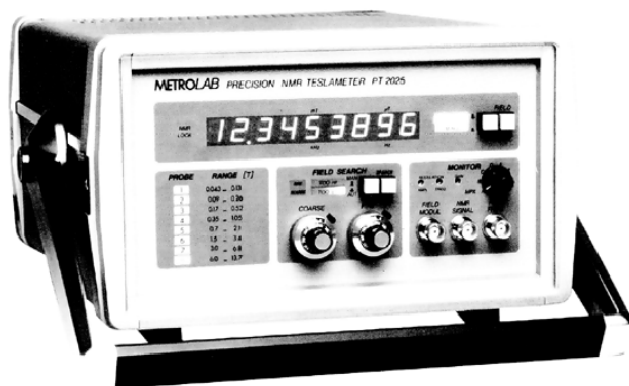
2. Connect the Preamp Box to the two Teslameter Probe inputs: the white Preamp Box wire to the Tesla-meter Probe left jack and the black Preamp Box wire to the right, lacing connector end of the Probe. See Illustration 1-12.
3. Turn on AC Power to the meter. The meter's rear view is shown in Illustrations 1-12 and 1-13.



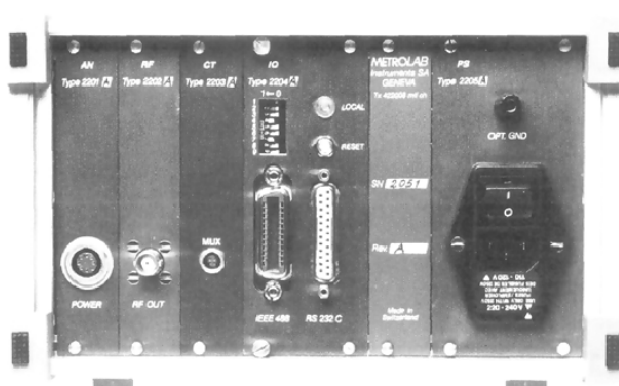
PROBE SET-UP AND TESLAMETER INTERCONNECTIONS

ILLUSTRATION 1-12

1-6-1 Set-Up (continued)



Front View



Rear View

TESLAMETER
ILLUSTRATION 1-13

4. Pull the "Lock / Manual" Switch and set in the "Manual" position.



If the Lock / Manual Switch is not in the "Manual" position at the start of field search during Ramping, the Teslameter's sweep will be in the $\pm 5,000$ gauss range and the system could lock onto the mechanical oscillation harmonic produced by the Cryo-cooler Coldhead, resulting in an erroneous reading.

5. Turn the Coarse and Fine Control Knobs fully counterclockwise (CCW).
6. Pull the "Freq / Field" Switch and set to the "Field" position. The Teslameter is now set up and ready to start monitoring the field when ramping commences.

Note:

The Teslameter will not "lock in" on a probe signal until magnetic field is approximately 4000 Gauss.

1-6-2 Teslameter Adjustment (Ramp Down)

1. Increase the Coarse Control Knob until a field reading of approximately 5.000 gauss is obtained. The field should be near but slightly below the actual field.
2. Slowly start increasing the Fine Control Knob while watching the "NO STROBE SIGNAL" LED. When the LED stops blinking and remains out, reposition the "Lock / Manual" Switch to the "Lock" position.

Note

If the signal will not lock on, change the position of the Probe Modulation Switch.

3. Now that the Teslameter is locked on the field. note that either the "LO" or "HI" LED is lighted. Turn the Coarse Control in the direction of whichever LED is on. Slowly it will go out and stay out.
4. As the magnetic field decreases, the probe sample resonant frequency will decrease below the present setting of the Teslameter. Therefore, decrease the setting of the Coarse Control Knob to keep the Teslameter locked on the probe sample.

Note:

The Fine Control does not function when the "Lock / Manual" Switch is in the "Locked" position.

■ 1-6-3 Teslameter Resynchronization

If the Teslameter should go out of sync while ramping the magnet up (or down), it can easily be resynchronized by the following procedure.

■ Manual Resynchronization

1. Reposition the "Lock / Manual" Switch to "MANUAL."

Note:

The "NO STROBE SIGNAL" LED will be on, and the "LO" and "HI" LEDs will be oscillating, indicating a search mode.

2. Note the present current reading on the Main Power Supply's Current Meter.
- 3. Multiply the current read from the Main Power Supply's Current Meter by 6.85. (There's approximately 6.85 gauss / amp.). Reset the meter at the resultant gauss level.
4. Slowly start increasing (if ramping up) the Coarse and / or Fine Control Knob while monitoring the "NO STROBE SIGNAL" LED.
5. Once the LED extinguishes, quickly place the Lock / Manual Switch in the "LOCK" position. The meter will now be in sync.

Note:

If the "HI" LED is lit, the Coarse Control Knob will have to be turned in the "HI" direction until the LED goes out. Repeat this adjustment as required until the parking field is reached.

1-6-3 Teslameter Resynchronization (continued)

Manual Resynchronization with Scope

An oscilloscope can be set up near the Teslameter to display and trigger on the "Field Modulation" signal from a jack on the Teslameter front panel. Adjust the time base to display one or two ramp wave forms. On the second channel display the "Signal" from the front panel of the Teslameter.

1. Leave the Teslameter in the "Locked" position.
2. Slowly turn the Coarse Control Knob in the direction the field is going (e.g., if ramping up, turn the knob up to the higher numbers).
3. As the meter is approaching the actual field, the baseline of the "FID" display will start to wander. Once the meter is in range of the field, the "FID" will appear on the oscilloscope trace as the meter locks on.
4. Once locked on, the "NO STROBE SIGNAL" LED will be out. Readjust the Coarse Control Knob slightly in the direction of the lighted "LO" / "HI" LED until that LED goes out.
5. Maintain tracking through end of the ramp sequence.

1-7 FIELD ADJUSTMENT / RAMPDOWN PREPARATION - PERSON 1**WARNING!**

MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THE ODORLESS, INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE WILL DISPLACE OXYGEN AND CAUSE ASPHYXIATION UNLESS SAFELY VENTED AWAY FROM THE MAGNET ROOM.

SECURE THE MAGNET ROOM DOORS IN THE OPEN POSITION BEFORE CHANGING MAGNETIC FIELD.

REVIEW AND FOLLOW SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.

1. Set up the field monitoring equipment, Probe and Teslameter in conformance with Section 1-6, Teslameter / Probe Set-Up And Adjustment. Make sure the Probe is positioned at the center of the magnet.
2. Make sure the Magnet Power Supply is installed, checked and adjusted in conformance with FUNCTIONAL CHECKS, Section 8, ESS7.5-1000-2-D-1236 Main Power Supply Calibration / Check. Verify that Heater 1 Main Power Supply is set for 810 ± 10 mA.
3. Make sure the Input Power Cable for the Power Supply is disconnected.
4. Make sure the Ramp Power Supply is connected to the magnet in conformance with SET-UP AND CALIBRATION, Sections 1-3, 1-4, and 1-5. Also make sure the Ramp Lead Hold-Down Tool is firmly in place and cooling gas is exiting from Main Lead Extensions.
5. Retrieve the Main Coil connection polarity in DATA SHEETS, Table 5-1, Magnet Ramping / Parking Current Log.
6. Set the Heater 1 Main and Heater 2 Shim Axial switches to the "OFF" position. Set the Current Adjust and Voltage Adjust Controls to "O" (full CCW).
7. Connect the Input Power Cable to the Service Main Power Supply.
8. Make sure a Digital Voltmeter (DVM) is connected to the end of the Voltage Sense Leads and terminated at the power supply. The voltmeter will measure voltage across the Lead Extensions of the magnet.
9. Disconnect and remove the Power Supply Jumper Cable from the power supply while wearing non-absorbant leather gloves with NO holes / tears. See Illustrations 1-6 and 1-7.

Note

Resistance measurements cannot be made and the magnet cannot be ramped with the Power Supply Jumper Cable connected.

1-8 RESISTANCE CHECKS - PERSON 1**Note**

CCW = counterclockwise -----> decrease

CW = clockwise -----> increase

1. Make sure the Current Adjust and Voltage Adjust Controls on the Main Power Supply are off (full CCW). Turn on input power to the Power Supply.
2. Set the Heater 2 Shim Axial Switch to the "I" (on) position to enable power supply output.
3. Set the Voltage Adjust Control to "400" (4 turns clockwise from zero) This will set a maximum limit of 3 V (~ 0.75 V / turn) for the power supply output.
4. Observe the Main Power Supply Ammeter and slowly turn the Current Adjust Control (clockwise = increase) to set 750A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
5. Record the voltage reading on the Digital Voltmeter (DVM): _____ mV.
 - A voltage reading < 50 mV indicates the Power Supply Jumper Cable is still in place.
 - A 50 - 150 mV reading indicates acceptable internal contact resistance of the Lead Extensions.
 - If the voltage reading is > 150 mV, turn the Current Adjust Control off (full CCW), turn off the Power Supply Input Power and reseal the Lead Extensions on the top of the magnet.
6. Repeat Steps 1 through 5 if the Lead Extensions were reseated. Repeated readings > 150 mV indicate the need to replace the Lead Extension Contact Bands in conformance with REPLACEMENT / MAINTENANCE, Section 8, Main Lead Extension Contact Band Replacement. Upon passing the internal resistance check, continue with Step 7.
7. Set the Toggle Switch on the Power Supply Voltmeter to the "Main Power Supply" position.
8. Read the voltage across the Power Supply Output Lugs. A Power Supply output voltage of 2.1 V or less at 750A indicates acceptable system resistance. If the voltage exceeds 2.1V during the test:
 - a. Turn the Current Adjust Control off (full CCW).
 - b. Turn off the input power.
 - c. Check and retighten the bolts securing the Main Power Cables to the Power Supply and Lead Extensions.
 - d. Turn the Power Supply back on.
 - e. Repeat Step 8.
9. Upon passing Step 8, turn the Current Adjust and Voltage Adjust Controls to zero (full CCW) and continue with Section 1-9, Magnet Field Adjustment / Rampdown - Person 1.

1-9 MAGNET FIELD ADJUSTMENT / RAMPDOWN - PERSON 1**WARNING!**

- **MAKE SURE THE MAGNET IS 98% FULL OF LIQUID HELIUM AND A FULL 250-LITER OR LARGER DEWAR OF LIQUID HELIUM IS CONNECTED TO THE TRANSFER LINE BEFORE RAMPING TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING DURING RAMPING TO A LEVEL WHERE A QUENCH MAY OCCUR.**
- **A RAMPING VOLTAGE > 2.5V AT THE LEAD EXTENSIONS OF THE MAGNET MAY CAUSE THE MAGNET TO QUENCH.**
- **MAKE SURE THE CONNECTION POLARITY AND THE CURRENT SETTINGS ARE THE SAME AS THE LAST ENTRY IN DATA SHEETS, TABLE 5-1, MAGNET RAMPING / PARKING CURRENT LOG. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.**
- **IF A QUENCH OCCURS DURING FIELD ADJUSTMENT / RAMPDOWN, IMMEDIATELY TURN THE “VOLTAGE ADJUST” AND “CURRENT ADJUST” CONTROLS OF THE MAIN COIL POWER SUPPLY TO ZERO (FULLY CCW). A QUENCH IS A RAPID DISCHARGE OF THE MAGNETIC FIELD WHICH WILL RESULT IN THE RAPID GENERATION AND EXPULSION OF HELIUM GAS, RUPTURING THE BURST DISC IN THE VENT SYSTEM.**

1-9-1 Magnet Field Adjustment - Person 1**1. Make sure:**

- A new (full) 250 liter Dewar is installed in conformance with SET-UP AND CALIBRATION, Section 3-6, Changing Helium Dewars.
- The Current Adjust and Voltage Adjust Controls are set to zero (full CCW).
- The Main Switch Heater is set to “OFF.”
- Magnet polarity and ramping current are retrieved from DATA SHEETS, Table 5-1, Magnet Ramping / Parking Current Log.

2. Turn the Power Supply and Auxiliary Switch Heater on.**3. Engage the Set Point Button (depress and hold until current is set) and set the Current Adjust Control for a maximum current of 10% above parking current.****4. Leave the Current Adjust Controls at setting in Step 3. Turn the toggle switch on the Power Supply Voltmeter to the “Main Coil” position.**

1-9-1 Magnet Field Adjustment - Person 1 (continued)

5. Use the Voltage Adjust Control to set the Power Supply output current to the parking current value.
6. Turn the Heater 1 Main Switch on the Power Supply to "ON" and wait 10 minutes for the switch to go normal (resistive).
7. Turn the Current Adjust Control to obtain the desired field as read on the Teslameter.
8. Adjust the magnetic field within the 29.803 - 29.811 MHz (6998.0 - 7002.0 gauss) range.
9. Maintain the current at the final setting until the Main switch goes into "persistent" mode.
10. Set Heater 1 Main Switch to "O" (off). Wait a minimum of 10 minutes for the switch to fully cool and go "persistent."

Note

Read the voltage on the Power Supply Voltmeter with its toggle switch in "Main Coil" position. When the switch goes "persistent" the voltage across the magnet terminals will drop to approximately 0.00V. This sudden voltage drop will indicate that the switch has gone into the persistent mode.

11. Record current value at which the switch went "persistent" in DATA SHEETS, Table 5-1, Magnet Ramping / Parking Current Log.
12. After the switch goes "persistent" slowly turn the Power Supply's Current Adjust Controls to zero (full CCW) over a one minute period. Make sure the voltage reading remains at zero.
13. Gradually turn the Voltage Adjust Control to zero over a one minute period and then turn the Power Supply off.
14. Turn off the Auxiliary Switch Heater and Ramp Power Supply.
15. Disconnect Input Power Cable from the Main Power Supply.
16. Slowly open Valve V2 while observing the Pressure Gauge to depressurize the Cryostat. When the pressure reaches 0.5 psig, close Valve V2.
17. Reconnect the Power Supply Jumper Cable's twist connectors to the power supply while wearing non-absorbant leather gloves with NO holes / tears. See Illustrations 1-5 and 1-7.
18. Remove the Ramp Lead Hold-Down Tool and replace the bolts on the Lead Assembly Mounting Plate.

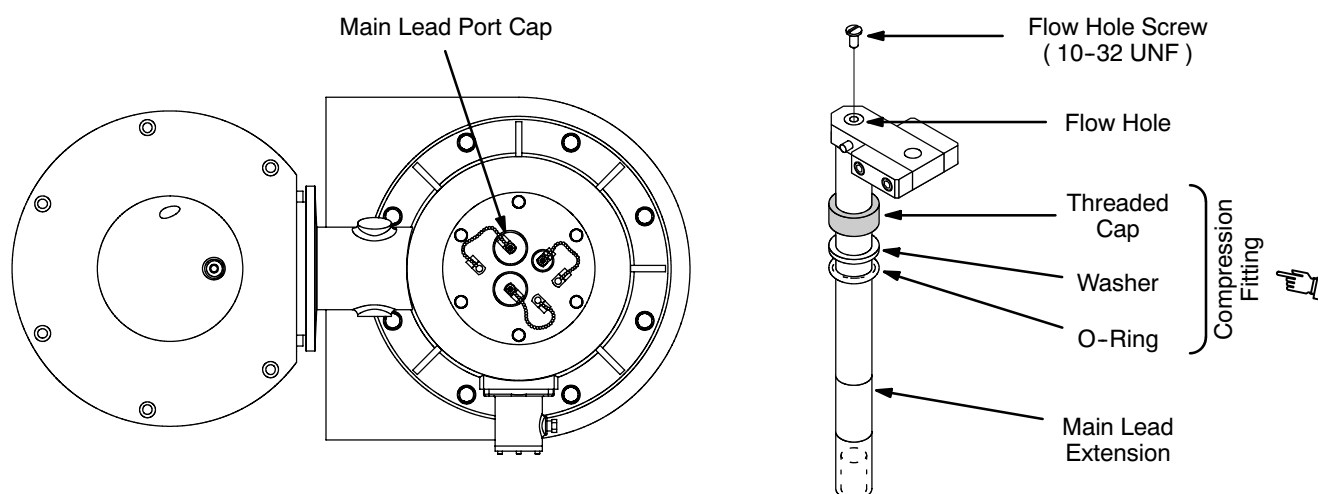
1-9-1 Magnet Field Adjustment - Person 1 (continued)**WARNING!**

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **MAKE ALL PRE-RAMP POWER SUPPLY PROTECTIVE CIRCUIT CONNECTIONS IN CONFORMANCE WITH SECTION 1-4, PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1.**
- **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL BEFORE CONTACTING THE MAIN LEAD PINS.**
- **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
- **WEAR NON-ABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
- **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**

19. After parking the magnet and disconnecting the Main Power Supply, plug and remove one Main Lead Extension at a time in the following sequence.
 - a. Open Valve V2 to depressurize the cryostat to < 0.5 psig. Close V2.
 - b. Install the screw into the flow hole of only one Main Lead Extension at a time. See Illustration 1-14.
 - c. Remove all ice around the Main Lead Port compression fitting on the Main Lead Extension that is being removed (i.e. the Main Lead Extension that has the flow hole plugged in Step 19b).
 - d. Unscrew the Main Lead Port compression fitting and remove the Main Lead Extension from the magnet. Immediately replace the cap onto the Main Lead Port.
 - e. Repeat these steps (19a - 19d) for the other Main Lead Extension.
20. Remove the Power Supply Jumper and Grounding Cables.

1-9-1 Magnet Field Adjustment - Person 1 (continued)

MAIN LEAD EXTENSION AND MAIN LEAD PORT COMPRESSION NUT
ILLUSTRATION 1-14

1-9-2 Magnet Rampdown to Zero Field - Person 1

1. Make sure:
 - The Current Adjust and Voltage Adjust Controls are set to zero (full CCW).
 - The Main Switch Heater is set to "OFF."
 - Magnet polarity and ramping current are retrieved from DATA SHEETS, Table 5-1, Magnet Ramping / Parking Current Log.
2. Turn the Power Supply and Auxiliary Switch Heater on.
3. Engage the Set Point Button (depress and hold until current is set) and set the Current Adjust Control for a maximum current of 10% above parking current.
4. Leave the Current Adjust Control at its setting; turn the power supply Voltmeter's switch to "MAIN COIL."
5. Use the Voltage Adjust Control to set the power supply output current to the parking current value.
6. Turn the Heater 1 Main Switch on the Power Supply to "ON" and wait 5 minutes for the switch to go normal (resistive).

1-9-2 Magnet Rampdown to Zero Field - Person 1 (continued)

7. Ramp down the magnet in conformance with Table 1-2 as follows:
- Slowly turn the power supply's Voltage Adjust Control to obtain the Main Coil Voltage value shown in the table, starting with -0.30 volts.

Note

Turn the power supply's Voltage Adjust Control counter clockwise to set a negative voltage value.

- Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Adjust Control to obtain the Main Coil Voltage for the next range.

TABLE 1-2
RAMPDOWN PROFILE

"Stage"	Main Coil (Ramp) Current Range (Amps)		Main Coil (Ramp) Voltage (Volts)
	From	TO	
1		750	-0.30
2	750	720	-0.35
3	720	700	-0.40
4	700	675	-0.50
5	675	650	-0.60
6	650	625	-0.80
7	625	600	-1.00
8	600	550	-1.30
9	550	500	-1.50
10	500	0	Gradually turn the Voltage Adjust Control to zero (full CCW), then turn the Current Adjust Control to zero (full CCW)

- When the ramping current reaches 500 amps, slowly turn the Voltage Adjust Control to 0.0 volts (full CCW) and then the Current Adjust Control to 0.0 amps (full CCW).
- Observe the voltmeter reading with the power supply's Voltmeter Select Switch in the "MAIN COIL" position. A zero reading on the power supply's digital Voltage and Current Meters indicates that the magnet is fully discharged.
- When the magnet is full discharged, set the Heater 1 Main and the Heater 2 Auxiliary Switches to "O" (off).
- Set the Main Power and Power On Switches to "OFF."
- Disconnect the input power cable from Magnet Power Supply.

1-9-2 Magnet Rampdown to Zero Field - Person 1 (continued)

13. Slowly open Valve V2 while observing the Pressure Gauge to depressurize the Cryostat. When the pressure reaches 0.5 psig, close Valve V2.
14. Reconnect the Power Supply Jumper Cable's twist connectors to the power supply while wearing non-absorbant leather gloves with NO holes / tears. See Illustrations 1-5 and 1-7.
15. Remove the Ramp Lead Hold-Down Tool and replace the bolts on the Lead Assembly Mounting Plate.
16. Make sure the magnet is at 98% helium fill level.

WARNING!

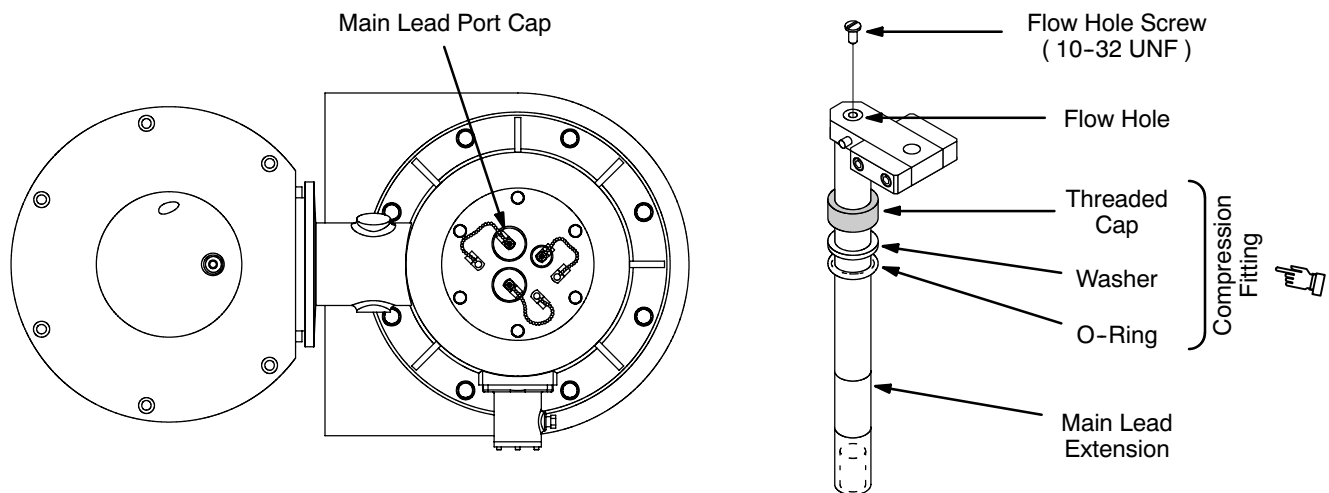
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 - **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL BEFORE CONTACTING THE MAIN LEAD PINS.**
 - **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
 - **WEAR NON-ABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
 - **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
17. After parking the magnet and disconnecting the Main Power Supply, plug and remove one Ramp Lead Extension at a time in the following sequence.
 - a. Open Valve V2 to depressurize the cryostat to < 0.5 psig. Close V2.
 - b. Install the screw into the flow hole of only one Main Lead Extension at a time. See Illustration 1-15.
 - c. Remove all ice around the Main Lead Port compression nut on the Main Lead Extension that is being removed (i.e. the Main Lead Extension that has the flow hole plugged in Step 17b).

1-9-2 Magnet Rampdown to Zero Field - Person 1 (continued)**17. (continued)**

- d. Unscrew the Main Lead Port compression fitting and remove the Main Lead Extension from the magnet. Immediately replace the cap onto the Main Lead Port.
- e. Repeat these steps (a - d) for the other Main Lead Extension.

18. Remove the Power Supply Jumper and Grounding Cables.

MAIN LEAD EXTENSION AND MAIN LEAD PORT COMPRESSION NUT
ILLUSTRATION 1-15